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METHODS, SYSTEMS, AND APPARATUSES FOR PASSIVE OPTICAL NETWORKS

Abstract

In various embodiments, the present disclosure includes a system for sending 50 gigabits per second (Gbps), 75 Gbps, and 100 Gbps at 50 gigabaud (GBaud) for passive optical networks (PON) downstream and upstream. The system allows for transmission of three data rates at a single baud-rate while only using 2-bits of information per sample. A motivation for sending three data rates at a single baud-rate is to allow for further granularity in the control of the data-rates for downstream and upstream traffic in a flexible PON system based on the link margin. For example, the system can use non-return-to-zero (NRZ) at 50 GBaud for 50 Gbps and can use four-level pulse-amplitude modulation (PAM-4) at 50 GBaud for 100 Gbps. In addition for 75 Gbps, a double square-8 (DSQ-8) constellation can be used at 50 GBaud.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 18/101,538 filed Jan. 25, 2023 entitled “Methods, Systems, and Apparatuses for Passive Optical Networks,” by Bart ZEYDEL, which is hereby incorporated by reference. [0002] The U.S. patent application Ser. No. 18/101,538 claims the benefit of U.S. Provisional Patent Application No. 63/303,412 filed on Jan. 26, 2022 entitled “System for Transmitting 3 Data Rates at Fixed Baud-Rate Using 2-Bits Per Sample for Passive Optical Networks,” by Bart ZEYDEL, which is hereby incorporated by reference. [0003] The U.S. patent application Ser. No. 18/101,538 claims the benefit of U.S. Provisional Patent Application No. 63/303,476 filed on Jan. 26, 2022 entitled “Receiver Supporting Multiple Interleaved Downstream Data Rates at Fixed Baud-Rate for Flexible Passive Optical Networks,” by Bart ZEYDEL, which is hereby incorporated by reference.

BACKGROUND

[0004] Passive optical networks (PON) utilize a downstream broadcast from a single terminal to multiple terminals. In traditional PON systems the downstream rate is fixed (for a given wavelength) and the upstream traffic is received in time interleaved bursts from the endpoints at multiple rates that are an integer multiple of the downstream traffic. For example, FIG. 1 shows an ITU-T 50-Gigabit-capable passive optical networks (50G-PON) system where the downstream rate is 50 gigabits per second (Gbps) and upstream rate for bursts can be either 50 Gbps, 25 Gbps, or 12.5 Gbps. For both upstream and downstream traffic to be placed on the optical distribution network (ODN), the data is provided to the optical line termination or optical line terminal (OLT) and optical network unit (ONU) using as few bits as possible. For 50 Gbps non-return-to-zero (NRZ), this requires 1-bit of information per sample. However, it is desirable to have a higher data rate for upstream and downstream traffic to improve the overall customer experience with the optical distribution network.

[0005] A proposal was made to support 50 gigabaud (GBaud) NRZ being defined in the ITU-T 50G-PON system and 100 Gbps 50 GBaud four-level pulse-amplitude modulation (PAM-4), which allows for a system to alternate between transmitting 50 Gbps and 100 Gbps for the downstream link using 2-bits of information per sample. Also proposed was the use of 100 Gbps at 50 GBaud in the upstream direction as an option for the upstream burst from an optical network terminal (ONT) or optical network unit (ONU). The two rates are achieved as follows: 50 Gbps is achieved using 50 GBaud NRZ; and 100 Gbps is achieved using 50 GBaud PAM-4 (4-level modulation per sample). The reason for using the different rates is to allow for links that have additional link margin to operate at a higher data rate. For example, a link that is closer to the OLT (has less optical power loss) could operate at 100 Gbps, while a link that is farther away and has more optical power loss would need to operate at 50 Gbps. To achieve the desired signal-to-noise ratio (SNR) improvement when sending NRZ compared to PAM-4, the outer modulation amplitude (OMA) of NRZ and PAM-4 are the same. FIG. 2 shows two-dimensional (2D) 2-bit constellation diagrams **202** and **204** for NRZ and PAM-4, respectively, at 50 GBaud. Note that for both constellation diagrams **202** and **204**, the 1st symbol on the X-axis represents the first bit of information in time while the 2nd symbol on the Y-axis represents the second bit of information in time. As shown in FIG. 2, the worst case spacing between constellation points can be seen to be a

delta of 3 for NRZ and a delta of 1 for PAM-4. The spacing difference is what results in the SNR improvement which allows for NRZ to operate on links with more optical power loss (OPL) compared to PAM-4.

[0006] FIG. 3 shows a flexible PON with 50 Gbps and 100 Gbps downstream on a single wavelength and 100 Gbps (PAM-4), 50 Gbps (NRZ), 25 Gbps (NRZ), and 12.5 Gbps (NRZ) upstream bursts. In this system of FIG. 3 each of the ONT receivers will operate in one of two scenarios. The first scenario is a link over which an ONT can only receive 50 Gbps traffic. The second scenario is a link where the ONT can receive PAM-4, and thus by definition also has sufficient SNR to receive NRZ. In all scenarios the ONT receiver can recover the bits of information transmitted at 50 Gbps. However, note that at least one disadvantage of the system of FIG. 3 is that the downstream rate drops by half when the 100 Gbps rate cannot be supported by one or more link margins of the system. In addition, the upstream bursts also drops by half when the 100 Gbps rate cannot be supported by one or more link margins of the system.

SUMMARY

[0007] Various embodiments in accordance with the present disclosure can address the disadvantages described above.

[0008] In various embodiments, the present disclosure includes a system for sending 50 gigabits per second (Gbps), 75 Gbps, and 100 Gbps at 50 gigabaud (GBaud) for passive optical networks (PON) downstream and upstream. The system allows for transmission of three data rates at a single baud-rate while only using 2-bits of information per sample. A motivation for sending three data rates at a single baud-rate is to allow for further granularity in the control of the data-rates for downstream and upstream traffic in a flexible PON system based on the link margin. For example, in various embodiments, the system can use non-return-to-zero (NRZ) at 50 GBaud for 50 Gbps and can use four-level pulse-amplitude modulation (PAM-4) at 50 GBaud for 100 Gbps. In addition for 75 Gbps, a double square-8 (DSQ-8) constellation can be used at 50 GBaud. This constellation allows for 1.5 bits to be sent per sample, by encoding 3-bits of information into two successive 2-bit samples. In various embodiments, note that this system of the present disclosure is not adding significant complexity to a system that only sends 50 Gbps and 100 Gbps at 50 GBaud for a flexible PON downstream and upstream. In various embodiments, the system is basically using similar components for the most part that would be used for a system that only sends 50 Gbps and 100 Gbps at 50 GBaud for flexible PON downstream and upstream with some slight additional processing and adds another rate (e.g., 75 Gbps) transparently to the transmitter with little overhead to the receiver (e.g., as described herein) and without latency overhead. In this manner, faster reception of data upstream and downstream can be provided in a flexible PON in various embodiments.

[0009] While various embodiments in accordance with the present disclosure have been specifically described within this Summary, it is noted that the claimed subject matter are not limited in any way by these various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and form a part of this specification and in which like numerals depict like elements, illustrate embodiments of the present disclosure and, together with the detailed description, serve to explain the principles of the disclosure. The drawings are not necessarily to scale.

[0011] FIG. 1 shows an ITU-T 50-Gigabit-capable passive optical networks (50 Gbps PON) system.

[0012] FIG. 2 shows two-dimensional (2D) 2-bit constellation diagrams for both non-return-to-zero

(NRZ) and four-level pulse-amplitude modulation (PAM-4) at 50 GBaud.

[0013] FIG. 3 shows a flexible PON with 50 Gbps and 100 Gbps downstream on a single wavelength and 100 Gbps (PAM-4), 50 Gbps (NRZ), 25 Gbps (NRZ), and 12.5 Gbps (NRZ) upstream bursts.

[0014] FIG. 4 shows 2D 2-bit constellation diagrams for 100 Gbps (PAM-4), 75 Gbps (double square-8 (DSQ-8)), and 50 Gbps (NRZ) at 50 GBaud in accordance with various embodiments of the present disclosure.

[0015] FIG. 5 shows a 2D 2-bit constellation diagram in accordance with various embodiments of the present disclosure.

[0016] FIG. 6 shows another 2D 2-bit constellation diagram in accordance with various embodiments of the present disclosure.

[0017] FIG. 7 shows 2D 2-bit constellation diagrams in accordance with various embodiments of the present disclosure.

[0018] FIG. 8 is a block diagram of a receiver supporting multiple interleaved downstream data rates at a fixed baud-rate for flexible passive optical networks in accordance with various embodiments of the present disclosure.

[0019] FIG. 9 is a method in accordance with various embodiments of the present disclosure.

[0020] FIG. 10 is a method in accordance with various embodiments of the present disclosure.

[0021] FIG. 11 is a block diagram of a PON system in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0022] Reference will now be made in detail to various embodiments in accordance with the present disclosure, examples of which are illustrated in the accompanying drawings. While described in conjunction with various embodiments, it will be understood that these various embodiments are not intended to limit the present disclosure. On the contrary, the present disclosure is intended to cover alternatives, modifications and equivalents, which may be included within the scope of the present disclosure. Furthermore, in the following detailed description of various embodiments in accordance with the present disclosure, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be evident to one of ordinary skill in the art that the present disclosure may be practiced without these specific details or with equivalents thereof. In other instances, well known methods, procedures, components, and circuits have not been described in detail as not to unnecessarily obscure aspects and features of the present disclosure.

[0023] Some portions of the detailed descriptions that follow are presented in terms of procedures and other representations of operations for systems, devices, and apparatuses like those disclosed herein. These descriptions and representations are the means used by those skilled in the art most effectively convey the substance of their work to others skilled in the art. In various embodiments of the present application, a procedure, operation, or the like, is conceived to be a self-consistent sequence of steps or instructions leading to a desired result. Operations described as separate blocks may be combined and performed in the same process step (that is, in the same time interval, after the preceding process step and before the next process step). Also, the operations may be performed in a different order than the order in which they are described herein. Furthermore, other processes and steps may be performed along with the processes and steps discussed herein; that is, there may be a number of process steps before, in between, and/or after the steps shown and described herein. Importantly, various embodiments according to the present disclosure can be implemented in conjunction with these other (perhaps conventional) processes and steps without significantly perturbing them. Generally speaking, various embodiments according to the present disclosure can replace portions of a conventional process without significantly affecting peripheral processes and steps.

[0024] It should be borne in mind, however, that all of these and similar terms are to be associated

with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the present disclosure, discussions utilizing terms such as “transmitting,” “receiving,” “determining,” “identifying,” “implementing,” “controlling,” “inputting,” “operating,” “analyzing,” “generating,” “extracting,” “processing,” “acquiring,” “performing,” “producing,” “providing,” “communicating,” “storing,” “creating,” “computing,” “loading” or the like, refer to actions and processes of a computing system or similar electronic computing device or processor. The computing system or similar electronic computing device manipulates and transforms data represented as physical (electronic) quantities within the computing system memories, registers or other such information storage, transmission or display devices.

[0025] Portions of the detailed description that follow are presented and discussed in terms of one or more methods. Although steps and sequencing thereof are disclosed in figures herein describing the operations of the one or more methods, such steps and sequencing are exemplary. Any method is well suited to performing various other steps or variations of the steps recited and/or shown herein, and in a sequence other than that depicted and/or described herein.

[0026] Various embodiments described herein may be discussed in the general context of computer-executable instructions residing on some form of computer-readable storage medium, such as program modules, executed by one or more computers or other devices. By way of example, and not limitation, computer-readable storage media may comprise non-transitory computer storage media and communication media. Generally, program modules include routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types. The functionality of the program modules may be combined or distributed as desired in various embodiments.

[0027] Computer storage media includes, but is not limited to, volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable ROM (EEPROM), flash memory or other memory technology, compact disk ROM (CD-ROM), digital versatile disks (DVDs) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed to retrieve that information.

[0028] In various embodiments, the present disclosure can include a system for transmitting three data rates at a fixed baud-rate using 2-bits per sample for passive optical networks (PON). More specifically, in various embodiments, the present disclosure can include a system for sending 50 gigabits per second (Gbps), 75 Gbps, and 100 Gbps at 50 gigabaud (GBaud) for passive optical networks downstream and upstream. The system allows for transmission of three data rates at a single baud-rate while only using 2-bits of information per sample. A motivation for sending three data rates at a single baud-rate is to allow for further granularity in the control of the data-rates for downstream and upstream traffic in a flexible passive optical networks (PON) system based on the link margin. For example, in various embodiments, the system can use non-return-to-zero (NRZ) modulation at 50 GBaud for 50 Gbps and can use four-level pulse-amplitude modulation (PAM-4) at 50 GBaud for 100 Gbps. In addition for 75 Gbps, a double square-8 (DSQ-8) constellation can be used at 50 GBaud. This constellation allows for 1.5 bits to be sent per sample, by encoding 3-bits of information into two successive 2-bit samples.

[0029] FIG. 4 shows two-dimensional (2D) 2-bit constellation diagrams **406**, **404**, and **402** for 100 Gbps (PAM-4), 75 Gbps (double square-8 (DSQ-8)), and 50 Gbps (NRZ), respectively, at 50 GBaud in accordance with various embodiments of the present disclosure. It is noted that a system in accordance with various embodiments of the present disclosure can allow for only 2-bits per

sample to be used to achieve the three desired data rates (e.g., 50 Gbps, 75 Gbps, and 100 Gbps), with the desired increase in worst case constellation spacing which allows for operation over links with different optical power loss for whatever reasons. A motivation for sending three data rates at a single baud-rate is to allow for further granularity in the control of the data-rates for downstream and upstream traffic in a flexible PON system based on the link margin. For example, within constellation diagram **402** the 50 Gbps NRZ worst case constellation point spacing is 3 (e.g., vertical and horizontal delta equal to 3), within constellation diagram **404** the 75 Gbps DSQ-8 worst case spacing is the square root of 2 ($\sqrt{2}$) (e.g., diagonal delta equal to $\sqrt{2}$), and within constellation diagram **406** the 100 Gbps PAM-4 worst case spacing is 1 (e.g., vertical and horizontal delta equal to 1).

[0030] Note that a flexible PON system in accordance with various embodiments of the present disclosure has the ability to selectively increase the speed of a particular link based on optical power loss and link margin. For example, in various embodiments, for an ONT receiver there can be three scenarios that exist for this flexible PON system: 1) a link with ONT is capable of only receiving 50 Gbps; 2) a link with ONT is capable of only receiving 75 Gbps and 50 Gbps; and 3) a link with ONT is capable of receiving 100 Gbps, 75 Gbps, and 50 Gbps. Note that each of these scenarios is further described herein with reference to FIGS. 5, 6, and 7, respectively.

[0031] With reference to FIG. 4, in various embodiments, a system is able to dynamically switch between the constellations **402**, **404**, and **406** at a single baud-rate for downstream and upstream traffic in a flexible PON system based on the link margin. Specifically, in various embodiments, a system is able to dynamically switch between the NRZ constellation **402**, the DSQ-8 constellation **404**, and PAM-4 constellation **406** at a single baud-rate for downstream and upstream traffic in a flexible PON system based on the link margin. In this manner, this allows for further granularity in the control of the data-rates for downstream and upstream traffic in a flexible PON system based on the link margin.

[0032] Note that within the DSQ-8 constellation **404**, it depends on both the 1st symbol and 2nd symbol. In this manner, it is noted that the 1st symbol is restricted based on the 2nd symbol. For example, if the 1st symbol is transmitted as a 1 or a 3, then the 2nd symbol can only transmit as a 0 or a 2. Therefore, within the DSQ-8 constellation **404**, there is a restriction on the point square formed by points **408**, **410**, **412**, and **414** and the point square formed by points **416**, **418**, **420**, and **422** where they are paired up such that they cannot be sent without knowing the pair of the symbol. Whereas with the PAM-4 constellation **406**, a 0, 1, 2, or 3 can be sent as the 1st symbol and the 2nd symbol regardless of what was previously sent. Furthermore, with the NRZ constellation **402**, a 0 or a 3 can be sent as the 1st symbol and the 2nd symbol regardless of what was previously sent.

[0033] FIG. 5 shows a 2D 2-bit constellation diagram **500** in accordance with various embodiments of the present disclosure. For example, on a link where the ONU can only recover 50 Gbps NRZ, the receiver can implement an NRZ slicer as shown by the constellation diagram **500** to recover 50 Gbps traffic in accordance with various embodiments of the present disclosure. For example, in various embodiments, to determine whether a received signal or bit (e.g., the 1st symbol or 2nd symbol) of the 50 Gbps traffic is a zero or a 3, the NRZ slicer can determine if it is greater than 1.5 or less than 1.5 shown by dashed lines **502** and **504** of the constellation diagram **500**. Specifically, if the received signal or bit (e.g., the 1st symbol or 2nd symbol) of the 50 Gbps traffic is greater than or equal to 1.5 (as shown by dashed line **502** or **504**), the NRZ slicer can determine that is a 3. And if the received signal or bit of the 50 Gbps traffic is less than 1.5 (as shown by dashed line **502** or **504**), then the NRZ slicer can determine that is a zero.

[0034] FIG. 6 shows a 2D 2-bit constellation diagram **600** in accordance with various embodiments of the present disclosure. For example, on a link where the ONU can only recover 75 Gbps and 50 Gbps NRZ, the receiver can implement a DSQ-8 slicer as shown by constellation diagram **600** to recover 75 Gbps traffic and an NRZ slicer as shown in constellation diagram **500** (FIG. 5) to recover 50 Gbps traffic in accordance with various embodiments of the present disclosure. It is

noted that in various embodiments, to determine (or recover) a received signal (or symbol) of the 75 Gbps traffic in the DSQ-8 constellation diagram **600**, the DSQ-8 slicer uses both the pair of symbols. That is why the diagonals **602**, **604**, **606**, **608**, and **610** are included within the DSQ-8 constellation diagram **600**. For vertical and horizontal such as shown in the NRZ constellation diagram **500** (FIG. 5), the determination (or recovery) of the 50 Gbps traffic can be done by the NRZ slicer based on a single symbol in time. However, for the DSQ-8 constellation diagram **600**, the determination (or recovery) of the 75 Gbps traffic can be done by the DSQ-8 slicer based on both the pair of symbols (e.g., both the 1st and 2nd symbols) as indicated by the diagonals **602**, **604**, **606**, **608**, and **610**.

[0035] FIG. 7 shows 2D 2-bit constellation diagrams **702**, **704**, and **706** in accordance with various embodiments of the present disclosure. For example, on a link where the ONU can recover 100 Gbps (PAM-4) traffic, 75 Gbps (DSQ-8) traffic, and 50 Gbps (NRZ) traffic, the receiver can implement a PAM-4 slicer to recover data for all three rates and modulations. Specifically, the receiver can implement a PAM-4 slicer as shown by constellation diagram **702** to recover 100 Gbps (PAM-4) traffic utilizing vertical dashed lines **708**, **710**, and **712** or horizontal dashed lines **714**, **716**, and **718** in accordance with various embodiments of the present disclosure. In addition, the receiver can also implement the PAM-4 slicer as shown by constellation diagram **704** to recover 75 Gbps (DSQ-8) traffic utilizing vertical dashed lines **708**, **710**, and **712** or horizontal dashed lines **714**, **716**, and **718** in accordance with various embodiments of the present disclosure. Furthermore, the receiver can also implement the PAM-4 slicer as shown in constellation diagram **706** to recover 50 Gbps (NRZ) traffic utilizing vertical dashed lines **708**, **710**, and **712** or horizontal dashed lines **714**, **716**, and **718** in accordance with various embodiments of the present disclosure. It is noted that while the PAM-4 slicer is able to be utilized to recover 100 Gbps (PAM-4) traffic, 75 Gbps (DSQ-8) traffic, and 50 Gbps (NRZ) traffic in various embodiments, the recovery of the 75 Gbps (DSQ-8) traffic and 50 Gbps (NRZ) traffic can be of lower quality because the PAM-4 slicer is not specifically designed for DSQ-8 traffic and NRZ traffic.

[0036] In various embodiments, it is noted that on a link where the ONU can recover 100 Gbps (PAM-4) traffic, 75 Gbps (DSQ-8) traffic, and 50 Gbps (NRZ) traffic, the receiver can include and dynamically implement in parallel a PAM-4 slicer to recover 100 Gbps traffic, a DSQ-8 slicer to recover 75 Gbps traffic, and a NRZ slicer to recover 50 Gbps traffic. Specifically, in various embodiments the receiver can include and dynamically implement in parallel a PAM-4 slicer as shown by constellation diagram **702** (FIG. 7) to recover 100 Gbps (PAM-4) traffic, a DSQ-8 slicer as shown by constellation diagram **600** (FIG. 6) to recover 75 Gbps (DSQ-8) traffic, and a NRZ slicer as shown by constellation diagram **500** (FIG. 5) to recover 50 Gbps (NRZ) traffic.

[0037] In various embodiments, on a link where the ONU can recover 100 Gbps (PAM-4) traffic, 75 Gbps (DSQ-8) traffic, and 50 Gbps (NRZ) traffic, the receiver can be implemented to include in parallel a PAM-4 slicer, a DSQ-8 slicer, and a NRZ slicer that can be utilized to determine which data type is being received where the receiver examines the signal statistics output by the PAM-4 slicer, DSQ-8 slicer, and NRZ slicer. In addition, the receiver can determine the signal to noise ratio (SNR) of the bits that are being received by the receiver. Based on all this information, the receiver can make a determination on what type of traffic it is receiving and then dynamically use the appropriate slicer (e.g., PAM-4 slicer, DSQ-8 slicer, or NRZ slicer) for recovering that incoming traffic.

[0038] FIG. 8 is a block diagram of a receiver **800** capable of supporting multiple interleaved downstream data rates at a fixed baud-rate for flexible passive optical networks (PON) in accordance with various embodiments of the present disclosure. In various embodiments, the receiver **800** can support, but is not limited to, recovery of interleaved 50 Gbps and 100 Gbps at 50 GBaud or interleaved 50 Gbps, 75 Gbps, and 100 Gbps at 50 GBaud for passive optical network downstream systems.

[0039] In various embodiments, on a link where the ONU can recover 100 Gbps (PAM-4) traffic,

75 Gbps (DSQ-8) traffic, and 50 Gbps (NRZ) traffic, the receiver **800** can be implemented to include a PAM-4 slicer block **820**, a DSQ-8 slicer block **816**, and a NRZ slicer block **812** coupled in parallel that can be utilized by a data detect block **836** of the receiver **800** to determine which data type is being received by examining the signal statistics (e.g., PAM-4 data **832**, DSQ-8 data **828**, and NRZ data **824**) output by the PAM-4 slicer block **820**, the DSQ-8 slicer block **816**, and the NRZ slicer block **812**. In addition, the signal statistics from the slicers **812**, **816**, and **820** of the receiver **800** can determine the signal to noise ratio (SNR) of the bits that are being received by the receiver **800**. Based on this information and other information to be described herein, the data detect block **836** can make a determination on what type of traffic the receiver **800** is receiving and then dynamically use the appropriate slicer (e.g., PAM-4 slicer block **820**, DSQ-8 slicer block **816**, or NRZ slicer block **812**) for recovering that incoming traffic.

[0040] With reference to FIG. **8**, note that in order to make the determination described above in various embodiments, it is desirable that the receiver **800** first receives some data. Therefore, the receiver **800** can include configuration delay lines **840**, **842**, and **844** in order for the receiver **800** to accumulate some data and provide some time for the rate detect block **836** to determine what type of data the receiver **800** is receiving before any data **848** is output by the receiver **800**. Note that in various embodiments the detection happens on the boundaries. In various embodiments, the receiver **800** is able to dynamically switch between the constellations (e.g., 500, 600, and 702) to achieve continuous data flow.

[0041] In various embodiments, the receiver **800** can include, but is not limited to, an analog to digital converter (ADC) **804** that is coupled to receive an input signal **803** from an TIA (transimpedance amplifier) interface **802**. In addition, the receiver **800** can include an equalizer **808** that can be implemented with, but is not limited to, a Feed-Forward Equalizer (FFE) or a FFE together with a Decision Feedback Equalizer (DFE), and the receiver **800** can include an equalizer (EQ) least mean squares (LMS) engine **806** which is configured to adapt the taps based on the LMS slicer error **860** and input data **805** that it and the equalizer **808** are both are coupled to receive from the ADC **804**. In addition, the receiver **800** can include, but is not limited to, a timing recovery block **810**, which is configured to recover the clock frequency and phase from the signal **809** received from the equalizer **808** based on the slicer data (timing recovery (TR) data) **854** and slicer error (TR error) **866**.

[0042] The receiver **800** of FIG. **8** can further include, but is not limited to, parallel slicer blocks **812**, **816**, and **820** for each of the three potential data rates 50 Gbps NRZ, 75 Gbps DSQ-8, and 100 Gbps PAM-4, respectively. It is noted that each of the parallel slicer blocks **812**, **816**, and **820** are coupled to receive the signal **809** that is output by the equalizer **808**. In addition, the receiver **800** can further include, but is not limited to, parallel error slicer NRZ block **814**, error slicer DSQ-8 block **818**, and error slicer PAM-4 block **822** that are each coupled to receive the signal **809** that is output by the equalizer **808**. Furthermore, the receiver **800** can include, but is not limited to, a rate detect block **836** that can determine the rate of the received data **803** of the receiver **800** based on the data and/or error output by the slicers blocks **812**, **816**, and **820** and the error slicer blocks **814**, **818**, and **822**. Specifically, the rate detect block **836** can be coupled to receive the NRZ slicer data **824**, DSQ-8 slicer data **828**, and PAM-4 slicer data **832** from the slicer NRZ block **812**, slicer DSQ-8 block **816**, and slicer PAM-4 block **820**, respectively. In addition, the rate detect block **836** can be coupled to receive the NRZ error signal **826**, DSQ-8 error signal **830**, and PAM-4 error signal **834** from the error slicer NRZ block **814**, error slicer DSQ-8 block **818**, and error slicer PAM-4 block **822**, respectively. The receiver **800** can also include, but is not limited to, configurable delay lines **840**, **842**, and **844** that are each coupled to receive the NRZ slicer data **824**, DSQ-8 slicer data **828**, and PAM-4 slicer data **832**, respectively, from the slicer NRZ block **812**, slicer DSQ-8 block **816**, and slicer PAM-4 block **820**, respectively, to allow for rate detection and selection without glitching of traffic. Note that within the receiver **800**, the output(s) of each of the configurable delay lines **840**, **842**, and **844** can be coupled with input(s) of a multiplexer **850**. The multiplexer

850 is coupled to receive a data out select signal **846** from the rate detect block **836** that is configured to control what is output (e.g., data out **848**) from the multiplexer **850**. It is pointed out that the output data **848** is the output for the receiver **800**. In various embodiments, the output data **848** continues on to a switch or house effectively.

[0043] The receiver **800** can further include, but is not limited to, a multiplexer **856** that is coupled to receive NRZ slicer data **824**, DSQ-8 slicer data **828**, and PAM-4 slicer data **832** from the slicer NRZ block **812**, slicer DSQ-8 block **816**, and slicer PAM-4 block **820**, respectively. In addition, the multiplexer **856** can be coupled to receive a timing recovery (TR) data select signal **852** from the rate detect block **836** that causes the multiplexer **856** to output TR data **854** to the timing recovery block **810**. The receiver **800** can also include, but is not limited to, a multiplexer **868** that is coupled to receive NRZ error signal **826**, DSQ-8 error signal **830**, and PAM-4 error signal **834** from the error slicer NRZ block **814**, error slicer DSQ-8 block **818**, and error slicer PAM-4 block **822**, respectively. Moreover, the multiplexer **868** can also be coupled to receive the TR data select signal **852** from the rate detect block **836** that causes the multiplexer **868** to output TR error signal **866** to the timing recovery block **810**. The receiver **800** can further include, but is not limited to, a multiplexer **862** that is coupled to receive NRZ error signal **826**, DSQ-8 error signal **830**, and PAM-4 error signal **834** from the error slicer NRZ block **814**, error slicer DSQ-8 block **818**, and error slicer PAM-4 block **822**, respectively. Furthermore, the multiplexer **862** can also be coupled to receive the LMS select signal **858** from the rate detect block **836** that causes the multiplexer **862** to output LMS error signal **860** to the equalizer (EQ) LMS engine **806**.

[0044] In various embodiments, the rate detect block **836** of FIG. **8** can be configured in one of three ways, but is not limited to such, to determine the received traffic data rate of the signal **803**. For example the first configuration of the rate detect block **836** can be implemented based on the mean-square error (e.g., of error signals **826**, **830**, and **834** received from error slicer blocks **814**, **818**, and **822**, respectively) of each slicer (e.g., **812**, **816**, and **820**) averaged over a programmable window of up to, but not limited to, 1024 samples is used to determine the nature of the received data by evaluating the mean-squared error of the Rx data over a window of up to, but not limited to, 1024 samples. The second configuration of the rate detect block **836** can be implemented based on evaluating the statistics of the received data (e.g., of data signals **824**, **828**, and **832**) from each of the slicers (e.g., **812**, **816**, and **820**) over a configurable window of up to, but not limited to, 1024 samples. The third configuration of the rate detect block **836** can be implemented based on a combination of the mean-square error (e.g., of error signals **826**, **830**, and **834** received from error slicer blocks **814**, **818**, and **822**, respectively) of each slicer (e.g., **812**, **816**, and **820**) and statistics of the received data (e.g., of data signals **824**, **828**, and **832** received from slicers **812**, **816**, and **820**) over a configurable window of up to, but not limited to, 1024 samples. In various embodiments, it is noted that each of the first, second, and third configurations can be implemented to operate with any number of samples that is greater than or less than 1024 samples. In various embodiments, it is noted that each sample can add some latency to the operation of the receiver **800**. For example, in various embodiments, every 50 samples can add 1 nanosecond (ns) of latency. Given that, 1000 samples can add 20 ns of latency to the operation of the receiver **800**. Therefore, there can be a tradeoff on the amount of samples used by the rate detect block **836** of the receiver **800** against how much latency will be added to the operation of the rate detect block **836**.

[0045] Based on the determination from the rate select block **836** of the received traffic data rate of the signal **803**, the rate select block **836** can output the data out select signal **846** that is received by the multiplexer **850** thereby controlling the data output **848** signal that is selected from the configurable delay lines **840**, **842**, and **844**. The configurable delay lines **840**, **842**, and **844** are programmed based on the averaging window used for the rate detect block **836**. In various embodiments, it is noted that this select signal **846** can also be configured to be overridden to operate on a specific slicer output (e.g., **824**, **828**, and **832**) as selected via configuration.

[0046] Within FIG. **8**, in various embodiments, note that the LMS select signal **858** can be driven

by either the rate detect block **836** or by a configurable select signal. The LMS error select **858** is then used to select error (e.g., **826**, **830**, and **834** received from error slicer blocks **814**, **818**, and **822**, respectively) from the chosen slicer (e.g., **812**, **816**, and **820**). In various embodiments, the timing recovery block **810** has a unique select signal for the TR data **854** and TR error **864** to be used by the timing recovery block **810** that can be driven by either the rate detect block **836** (e.g., via the multiplexers **856** and **868** that each receive the TR select signal **852** from the rate detect block **836**) or by a configurable select signal.

[0047] In various embodiments, it is noted that the purpose of each of the slicer blocks **812**, **816**, and **820** is to determine what the value is of each bit it receives.

[0048] In various embodiments, the receiver **800** of FIG. **8** can include the ADC **804** that is coupled to receive the input signal **803** from the TIA interface **802**. The ADC **804** is coupled to provide an output signal **805** to the EQ LMS engine **806** and the equalizer **808**. The EQ LMS engine **806** is coupled to receive the LMS error **860** from multiplexer **862** and is also coupled to output a signal **807** to the equalizer **808**. The equalizer **808** is coupled to provide the slicer input signal **809** to the timing recovery block **810**, slicer NRZ block **812**, error slicer NRZ block **814**, slicer DSQ-8 block **816**, error slicer DSQ-8 block **818**, slicer PAM-4 block **820**, and error slicer PAM-4 block **822**. Note that the slicer NRZ block **812** is coupled to output the NRZ data **824** to the error slicer NRZ block **814**, the rate detect block **836**, the configurable delay **840**, and the multiplexer **856**.

Furthermore, the error slicer NRZ block **814** is coupled to output the NRZ error **826** to the rate detect block **836**, the multiplexer **862**, and the multiplexer **868**. The slicer DSQ-8 block **816** is coupled to output the DSQ-8 data **828** to the error slicer DSQ-8 block **818**, the rate detect block **836**, the configurable delay **842**, and the multiplexer **856**. Additionally, the error slicer DSQ-8 block **818** is coupled to output the DSQ-8 error **830** to the rate detect block **836**, the multiplexer **862**, and the multiplexer **868**. The slicer PAM-4 block **820** is coupled to output the PAM-4 data **832** to the error slicer PAM-4 block **822**, the rate detect block **836**, the configurable delay **844**, and the multiplexer **856**. Moreover, the error slicer PAM-4 block **822** is coupled to output the PAM-4 error **834** to the rate detect block **836**, the multiplexer **862**, and the multiplexer **868**.

[0049] In various embodiments, note that the configurable delay lines **840**, **842**, and **844** can be coupled to output the NRZ data **824**, DSQ-8 data **828**, and PAM-4 data **832**, respectively, to the multiplexer **850**. It is noted that the multiplexer **850** is coupled to receive the data out select **846** from the rate detect block **836** and note that the multiplexer **850** is also coupled to output the data out **848** for the receiver **800**. The multiplexer **856** is coupled to receive the TR select **852** from the rate detect block **836** and note that the multiplexer **856** is also coupled to output the TR data **854** to the timing recovery block **810**. The multiplexer **862** is coupled to receive the LMS select **858** from the rate detect block **836** and note that the multiplexer **862** is also coupled to output the LMS error **860** to the EQ LMS engine **806**. The multiplexer **868** is coupled to receive the TR select **852** from the rate detect block **836** and note that the multiplexer **868** is also coupled to output the TR error **866** to the timing recovery block **810**. It is noted that the rate detect block **836** is coupled to output the data out select **846** to multiplexer **850**. In addition, the rate detect block **836** is coupled to output the TR select **852** to both the multiplexers **856** and **868**. Moreover, the rate detect block **836** is coupled to output the LMS select **858** to the multiplexer **862**. Note that the timing recovery block **810** is coupled to output the signal(s) **811** to the ADC **804**.

[0050] It is noted that the receiver **800** may not include all of the elements illustrated by FIG. **8**. In addition, the receiver **800** can be implemented to include one or more elements not illustrated by FIG. **8**. It is pointed out that the receiver **800** can be utilized or implemented in any manner similar to that described and/or shown herein, but is not limited to such.

[0051] FIG. **9** is a method **900** in accordance with various embodiments of the present disclosure.

[0052] At operation **902**, an input signal is received that is associated with a passive optical network. Note that operation **902** can be implemented in a wide variety of ways. For example, operation **902** can be implemented in any manner similar to that described and/or shown by the

present disclosure, but is not limited to such.

[0053] At operation **904** of FIG. **9**, the data rate of the input signal is identified. Note that operation **904** can be implemented in a wide variety of ways. For example, operation **904** can be implemented in any manner similar to that described and/or shown by the present disclosure, but is not limited to such.

[0054] At operation **906**, one of three data rates is transmitted at a fixed baud-rate using 2-bits per sample for a passive optical network, wherein the transmitting can be upstream or downstream within the passive optical network. It is noted that the three data rates comprise a first data rate at a first modulation, a second data rate at a second modulation, and a third data rate at a third modulation. In various embodiments, the first rate comprises 50 gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation, the second rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-8) modulation, and the third rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4). In various embodiments, it is pointed out that operation **906** can be performed dynamically based on the identified data rate of the input signal. Note that operation **906** can be implemented in a wide variety of ways. For example, operation **906** can be implemented in any manner similar to that described and/or shown by the present disclosure, but is not limited to such.

[0055] FIG. **10** is a method **1000** in accordance with various embodiments of the present disclosure.

[0056] At operation **1002**, a receiver (e.g., **800**) receives an input signal comprising one of multiple data rates at a fixed baud-rate for a flexible passive optical network. In various embodiments, the multiple data rates comprise a first data rate comprising a first modulation; a second data rate comprising a second modulation; and a third data rate comprising a third modulation where each of the first modulation, second modulation, and third modulation are different. In various embodiments, the first modulation comprises non-return-to-zero (NRZ) modulation, the second modulation comprises double square-8 (DSQ-8) modulation, and the third modulation comprises four-level pulse-amplitude modulation (PAM-4). In various embodiments, the first data rate comprises 50 gigabits per second (Gbps), the second data rate comprises 75 Gbps, and the third data rate comprises 100 Gbps. Note that operation **1002** can be implemented in a wide variety of ways. For example, operation **1002** can be implemented in any manner similar to that described and/or shown by the present disclosure, but is not limited to such.

[0057] At operation **1004** of FIG. **10**, the data rate is identified (or determined) of the input signal. Note that operation **1004** can be implemented in a wide variety of ways. For example, operation **1004** can be implemented in any manner similar to that described and/or shown by the present disclosure, but is not limited to such.

[0058] At operation **1006**, an output (e.g., **848**) of the receiver is dynamically controlled based on the identified data rate of the input signal. Note that operation **1006** can be implemented in a wide variety of ways. For example, operation **1006** can be implemented in any manner similar to that described and/or shown by the present disclosure, but is not limited to such.

[0059] FIG. **11** is a block diagram of a passive optical networks (PON) system **1100** in accordance with various embodiments of the present disclosure. In various embodiments, it is noted that any embodiment described and/or shown by the present disclosure can be implemented as part of (or within) the PON system **1100**, but is not limited to such. In various embodiments, the PON system **1100** can be for sending 50 Gbps, 75 Gbps, and 100 Gbps at 50 GBaud for PON downstream **1110** and upstream **1112**. More specifically, the system **1100** allows for transmission of three data rates at a single baud-rate while only using 2-bits of information per sample. Note that within system **1100**, a motivation for sending three data rates at a single baud-rate is to allow for further granularity in the control of the data-rates for downstream **1110** and upstream **1112** traffic in a flexible PON system **1100** based on the link margin. For example, in various embodiments, the system **1100** can

use NRZ at 50 GBaud for 50 Gbps, can use DSQ-8 at 50 GBaud for 75 Gbps, and can use PAM-4 at 50 GBaud for 100 Gbps. It is noted that utilizing the DSQ-8 constellation allows for 1.5 bits to be sent per sample, by encoding 3-bits of information into two successive 2-bit samples. In various embodiments, note that the system **1100** is not adding significant complexity to a system that only sends 50 Gbps and 100 Gbps at 50 GBaud for a flexible PON downstream and upstream. In various embodiments, the system **1100** can basically use similar components for the most part that would be used for a system that only sends 50 Gbps and 100 Gbps at 50 GBaud for flexible PON downstream and upstream with some slight additional processing and adds another rate (e.g., 75 Gbps) transparently to the transmitter with little overhead to the receiver (e.g., as described herein) and without latency overhead. In this manner, faster reception of data upstream **1112** and downstream **1110** can be provided in the flexible PON system **1100** in various embodiments.

[0060] In various embodiments, the PON system **1100** can include the Internet **1102** that can be coupled with an Internet Service Provider (ISP) **1104**. Note that the ISP **1104** can include one or more OLT switches (e.g., **1106**). In addition, each OLT switch **1106** can include one or more OLTs (e.g., **1108a**, **1108b**, and/or **1108c**). The OLT **1108b** can be coupled with a splitter (1:N) **1114**. It is noted that each of the OLTs **1108a** and **1108c** can be coupled with a splitter (1:N) similar to splitter **1114**. The splitter **1114** can be coupled with one or more ONTs (e.g., **1116a**, **1116b**, **1116c**, **1116d**, and/or **1116e**). In various embodiments, each of the ONTs **1116a**, **1116b**, **1116c**, **1116d**, and **1116e** can be associated with each of the buildings (or houses or structures or businesses) **1118a**, **1118b**, **1118c**, **1118d**, and **1118e**, respectively.

[0061] In various embodiments, the downstream traffic **1110** of the PON system **1100** of FIG. **11** can be implemented to include, but is not limited to, broadcasting at varying rates such as 50 Gbps (NRZ), 75 Gbps (DSQ-8), and 100 Gbps (PAM-4) at a single wavelength such as 50 GBaud. In this manner, each ONT (e.g., **1116a**, **1116b**, **1116c**, **1116d**, and **1116e**) can receive data traffic at varying rates such as 50 Gbps (NRZ), 75 Gbps (DSQ-8), and 100 Gbps (PAM-4) at a single wavelength such as 50 GBaud, but is not limited to such. In addition, in various embodiments, the upstream traffic **1112** of the system **1100** can be implemented to include, but is not limited to, interleaving bursts of traffic from each ONT (e.g., **1116a**, **1116b**, **1116c**, **1116d**, and **1116e**) such that the bursts can be either 50 Gbps (NRZ), 75 Gbps (DSQ-8), and 100 Gbps (PAM-4). In this manner, each OLT (e.g., **1108b**) can receive interleaved bursts of traffic at 50 Gbps (NRZ), 75 Gbps (DSQ-8), and 100 Gbps (PAM-4), but is not limited to such.

[0062] It is noted that the PON system **1100** may not include all of the elements illustrated by FIG. **11**. In addition, the system **1100** can be implemented to include one or more elements not illustrated by FIG. **11**. It is pointed out that the system **1100** can be utilized or implemented in any manner similar to that described and/or shown herein, but is not limited to such.

[0063] Note that the following are examples in accordance with various embodiments of the present disclosure.

[0064] Example 1. An optical network receiver comprising: [0065] a first rate slicer coupled to receive an input signal and configured to recover data at a first rate and a first modulation; [0066] a second rate slicer coupled to receive the input signal and configured to recover data at a second rate and a second modulation; [0067] a third rate slicer coupled to receive the input signal and configured to recover data at a third rate and a third modulation, wherein the first, second, and third rate slicers are coupled in parallel to receive the input signal; and [0068] a rate detect block coupled to receive an output from the first rate slicer, an output from the second rate slicer, and an output from the third rate slicer, wherein the rate detect block is configured to determine the rate of the input signal received by the optical network receiver.

[0069] Example 2. The optical network receiver of Example 1, wherein the rate detect block is configured to dynamically control an output signal of the optical network receiver based on the determined rate of the input signal.

[0070] Example 3. The optical network receiver of Example 1, wherein the first rate comprises 50

gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation and the second rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-8) modulation.

[0071] Example 4. The optical network receiver of Example 3, wherein the third rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4).

[0072] Example 5. The optical network receiver of Example 1, wherein the second rate comprises 75 Gbps and the second modulation comprises DSQ-8 modulation and the third rate comprises 100 Gbps and the third modulation comprises PAM-4.

[0073] Example 6. The optical network receiver of Example 1, wherein the input signal comprises a fixed baud-rate.

[0074] Example 7. The optical network receiver of Example 2, wherein the input signal comprises a fixed baud-rate.

[0075] Example 8. The optical network receiver of Example 4, wherein the input signal comprises a fixed baud-rate.

[0076] Example 9. The optical network receiver of Example 1, further comprising: [0077] a first rate error block coupled to receive the input signal and coupled to receive the output from the first rate slicer; [0078] a second rate error block coupled to receive the input signal and coupled to receive the output from the second rate slicer; and [0079] a third rate error block coupled to receive the input signal and coupled to receive the output from the third rate slicer, [0080] wherein the rate detect block coupled to receive an output from the first rate error block, the second rate error block, and the third rate error block.

[0081] Example 10. A method comprising: [0082] transmitting one of three data rates at a fixed baud-rate using 2-bits per sample for a passive optical network, wherein the transmitting can be upstream or downstream within the passive optical network; [0083] wherein the three data rates comprise a first data rate at a first modulation, a second data rate at a second modulation, and a third data rate at a third modulation.

[0084] Example 11. The method of Example 10, further comprising: [0085] receiving an input signal; and [0086] identifying the data rate of the input signal.

[0087] Example 12. The method of Example 11, wherein the transmitting is performed dynamically based on the identified data rate of the input signal.

[0088] Example 13. The method of Example 10, wherein the first rate comprises 50 gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation and the second rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-8) modulation.

[0089] Example 14. The method of Example 13, wherein the third rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4).

[0090] Example 15. The method of Example 10, wherein the second rate comprises 75 Gbps and the second modulation comprises DSQ-8 modulation and the third rate comprises 100 Gbps and the third modulation comprises PAM-4.

[0091] Example 16. A method comprising: [0092] receiving via a receiver an input signal comprising one of multiple data rates at a fixed baud-rate for a flexible passive optical network; [0093] identifying the data rate of the input signal; and [0094] dynamically controlling an output of receiver based on the identified data rate of the input signal.

[0095] Example 17. The method of Example 16, wherein the multiple data rates comprise: [0096] a first data rate comprising a first modulation; [0097] a second data rate comprising a second modulation; [0098] a third data rate comprising a third modulation; [0099] wherein each of the first modulation, second modulation, and third modulation are different.

[0100] Example 18. The method of Example 17, wherein the first modulation comprises non-return-to-zero (NRZ) modulation and the second modulation comprises double square-8 (DSQ-8) modulation.

[0101] Example 19. The method of Example 18, wherein the third modulation comprises four-level pulse-amplitude modulation (PAM-4).

[0102] Example 20. The method of Example 19, wherein the first data rate comprises 50 gigabits per second (Gbps), the second data rate comprises 75 Gbps, and the third data rate comprises 100 Gbps.

[0103] Various embodiments of the present disclosure are thus described. While the present disclosure has been described in particular embodiments, it should be appreciated that the present disclosure should not be construed as limited by such embodiments, but rather construed according to the following claims.

Claims

1. A method comprising: transmitting one of multiple data rates at a fixed baud-rate for a passive optical network, wherein the transmitting can be upstream or downstream within the passive optical network; wherein the multiple data rates comprise a first data rate at a first modulation, a second data rate at a second modulation, and a third data rate at a third modulation.
2. The method of claim 1, further comprising: receiving an input signal; and identifying the data rate of the input signal.
3. The method of claim 2, wherein the transmitting is performed dynamically based on the identified data rate of the input signal.
4. The method of claim 1, wherein the first data rate comprises 50 gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation and the second data rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-8) modulation.
5. The method of claim 4, wherein the third data rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4).
6. The method of claim 1, wherein the second data rate comprises 75 Gbps and the second modulation comprises DSQ-8 modulation and the third data rate comprises 100 Gbps and the third modulation comprises PAM-4.
7. The method of claim 1, wherein the first modulation comprises non-return-to-zero (NRZ) modulation and the second modulation comprises DSQ-8 modulation.
8. The method of claim 7, wherein the third modulation comprises PAM-4.
9. A passive optical network comprising: an optical line terminal; and an optical network terminal coupled with the optical line terminal; wherein the optical line terminal is capable of transmitting downstream to the optical network terminal one of multiple data rates at a fixed baud-rate for the passive optical network, wherein the multiple data rates comprise a first data rate at a first modulation, a second data rate at a second modulation, and a third data rate at a third modulation; wherein the optical network terminal is capable of transmitting upstream to the optical line terminal one of the multiple data rates at the fixed baud-rate for the passive optical network.
10. The passive optical network of claim 9, wherein the transmitting downstream comprises broadcasting.
11. The passive optical network of claim 9, wherein the transmitting upstream comprises interleaving bursts.
12. The passive optical network of claim 11, wherein the transmitting downstream comprises broadcasting.
13. The passive optical network of claim 9, wherein the optical network terminal is capable of receiving an input signal and identifying the data rate of the input signal; wherein the transmitting upstream is performed dynamically based on the identified data rate of the input signal.
14. The passive optical network of claim 9, wherein the first data rate comprises 50 gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation, the second data rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-

- 8) modulation, and the third data rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4).
- 15.** A method comprising: transmitting one of three data rates at a fixed baud-rate using 2-bits per sample for a passive optical network, wherein the transmitting can be upstream or downstream within the passive optical network; wherein the three data rates comprise a first data rate at a first modulation, a second data rate at a second modulation, and a third data rate at a third modulation.
- 16.** The method of claim 15, further comprising: receiving an input signal; and identifying the data rate of the input signal.
- 17.** The method of claim 16, wherein the transmitting is performed dynamically based on the identified data rate of the input signal.
- 18.** The method of claim 15, wherein the first data rate comprises 50 gigabits per second (Gbps) and the first modulation comprises non-return-to-zero (NRZ) modulation and the second data rate comprises 75 Gbps and the second modulation comprises double square-8 (DSQ-8) modulation.
- 19.** The method of claim 18, wherein the third data rate comprises 100 Gbps and the third modulation comprises four-level pulse-amplitude modulation (PAM-4).
- 20.** The method of claim 15, wherein the second data rate comprises 75 Gbps and the second modulation comprises DSQ-8 modulation and the third data rate comprises 100 Gbps and the third modulation comprises PAM-4.
-